

Conditional probability



1 $\frac{1}{3}$

2

a 0.23

b 0.73

c $\frac{27}{50}$

d $\frac{23}{56}$

e $\frac{17}{50}$

3

a $x = 0$

b $x = 0.2, 0.3$

c $x = 0.3$

4

a

	2	4	5
3	5	7	8
7	9	11	12

b $\frac{2}{5}$

c 1

5

a

	1	2	4	6
1	1	2	4	6
2	2	4	8	12
3	3	6	12	18
4	4	8	16	24
5	5	10	20	30
6	6	12	24	36

b $\frac{1}{2}$

c $\frac{3}{10}$

d $\frac{1}{2}$

Dependent and independent events

6

a Independent

b Dependent

c Dependent

d Dependent

7

a Yes



b No

8 0.2

9

a

i 0.3

iii 0.1

ii 0.2

iv Non-mutually exclusive

b

i 0.2

iii 0

ii 0.1

iv Mutually exclusive

c

i 0.6

iii 0.2

ii 0.1

iv Non-mutually exclusive

d

i 0.1

iii 0

ii 0.7

iv Mutually exclusive

10

a

i 0.42

ii 0.12

iii 0.18

iv 0.18

v Yes

b

i 0.15

ii 0.05

iii 0.05

iv 0.02

v No

11

a 0.8

b 0.05

c $P(B) = x + 0.05$ or $P(B) = \frac{x}{0.8}$

d $x = 0.2$

e 0.25

12

a 0.7

b Yes

c 0.5

13 $\frac{8}{33}$

14 a $\frac{32}{663}$



b $\frac{4}{39}$

15 a i $\frac{33}{16\,660}$

ii $\frac{1}{649\,740}$

b 2 598 960

16 a No

b $\frac{1}{6}$

17 a $\frac{1}{45}$

b $\frac{1}{44}$

c No

d Dependent

18 a $\frac{1}{51}$

b $\frac{1}{50}$

c Dependent

19 a $\frac{5}{18}$

b $\frac{1}{3}$

20 a $\frac{1}{64}$

b $\frac{11}{850}$

21 a $\frac{1}{3}$

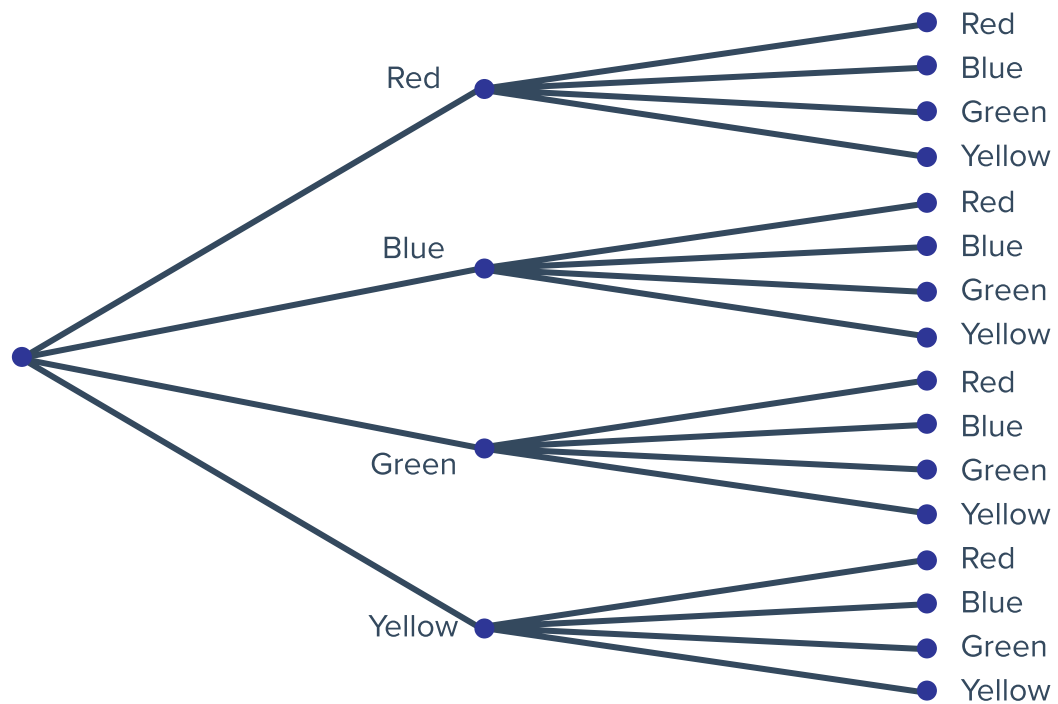
b $\frac{1}{12}$



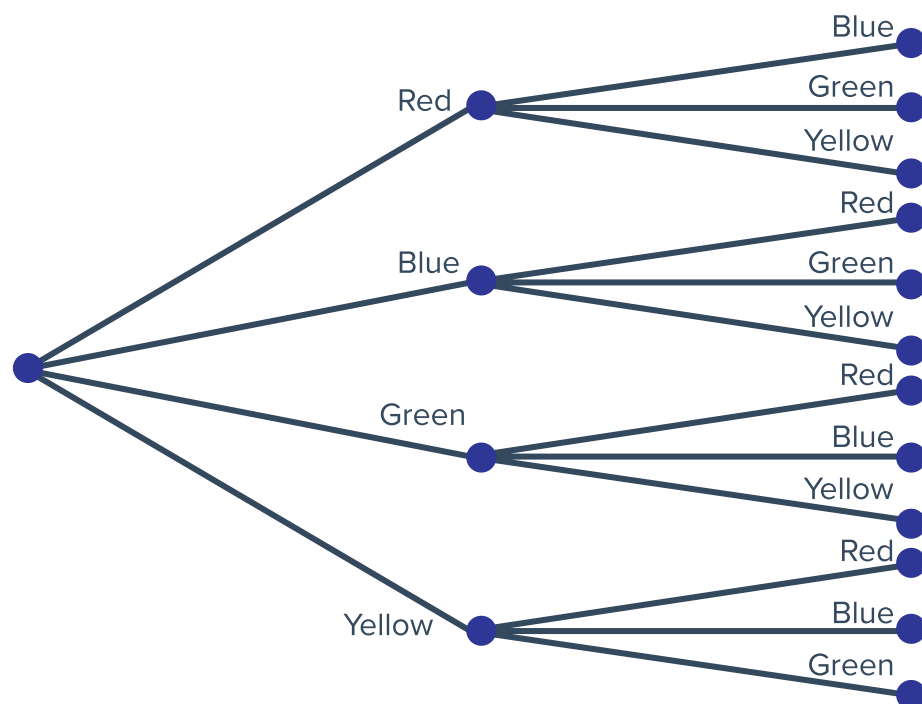
Tree diagrams and probability trees

22

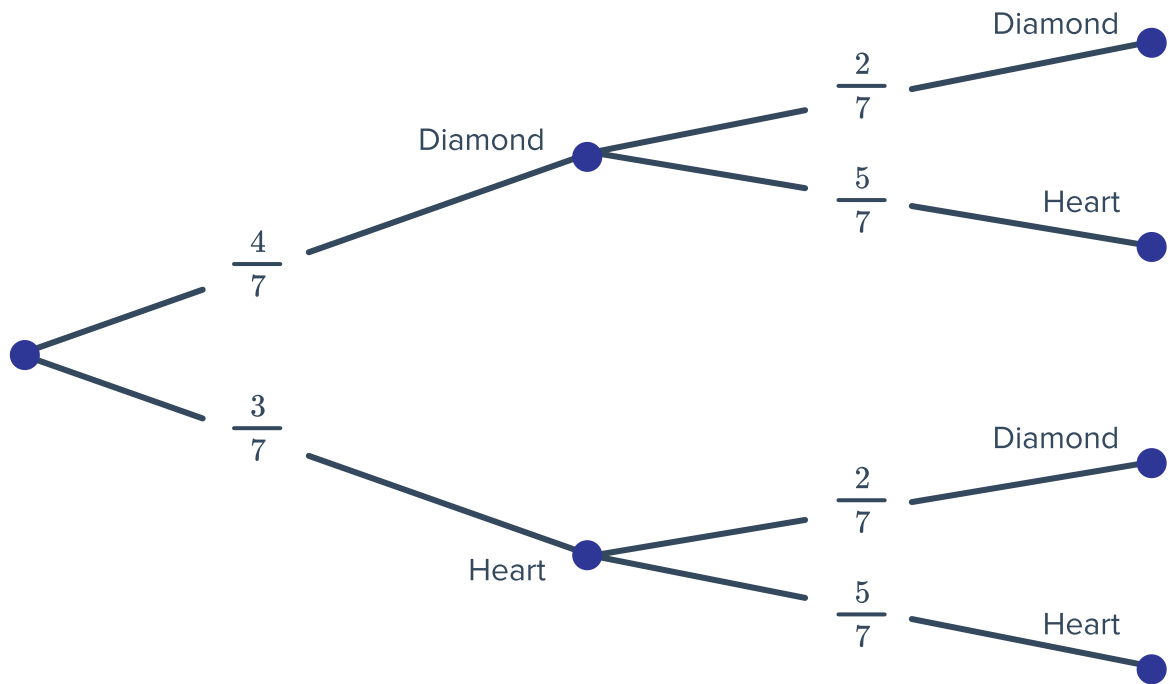
a



b

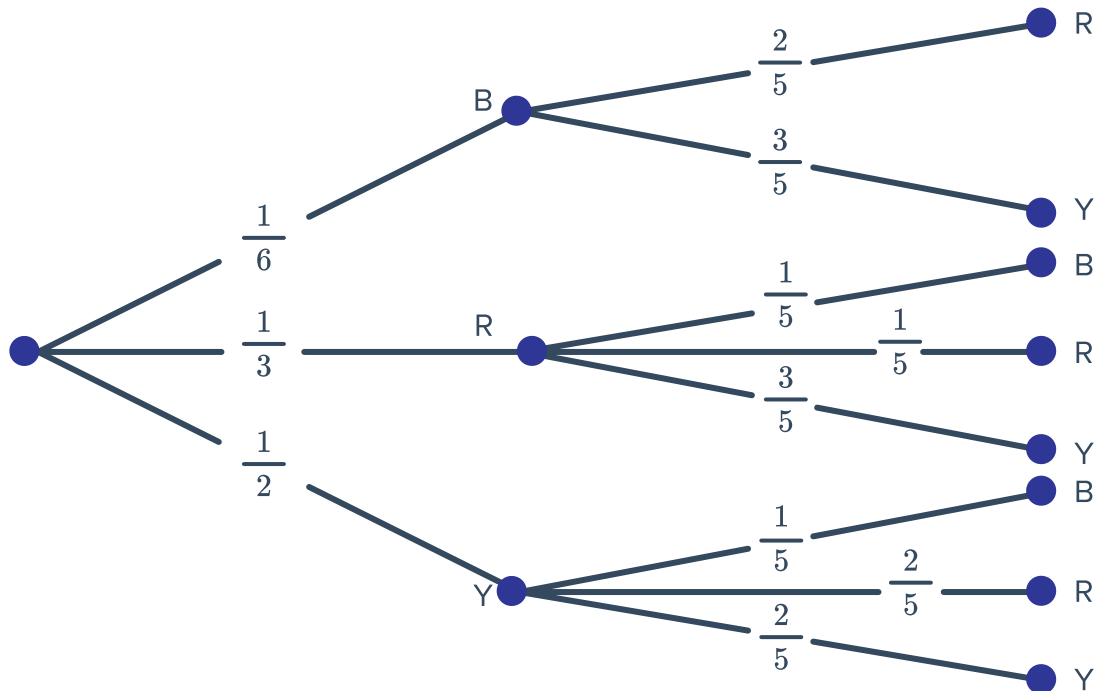


23
a



b $\frac{15}{49}$

24
a



b

i $\frac{1}{10}$
v 0

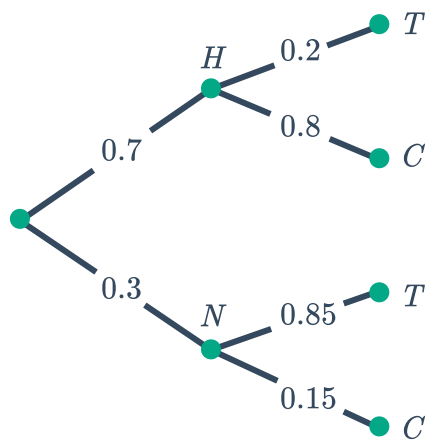
ii $\frac{1}{15}$
vi $\frac{1}{5}$

iii $\frac{1}{15}$
vii $\frac{2}{5}$

iv $\frac{1}{5}$

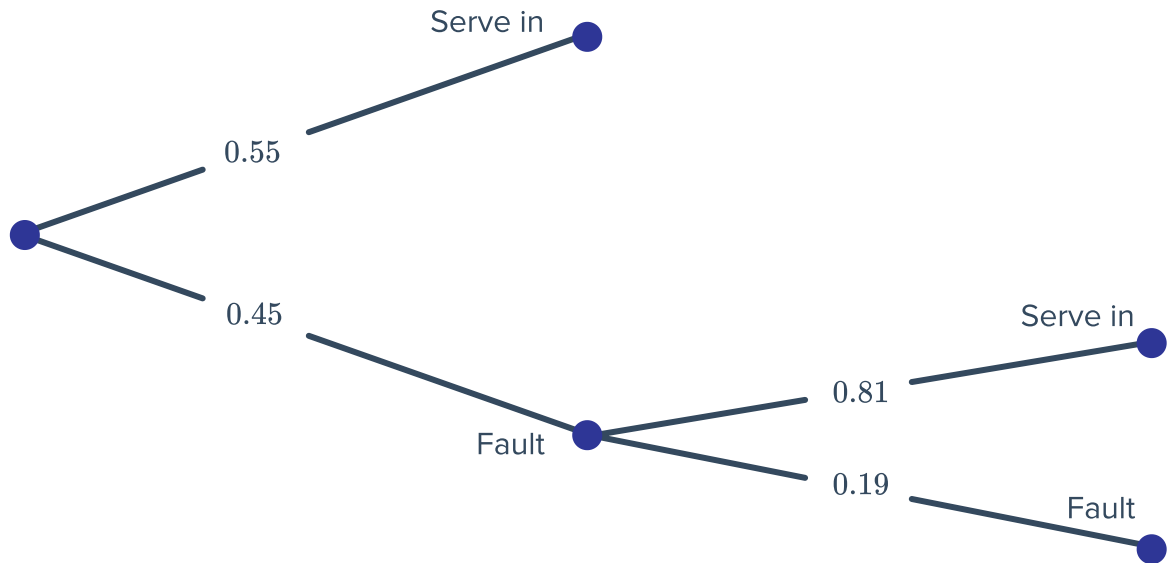


b 0.35



27

a

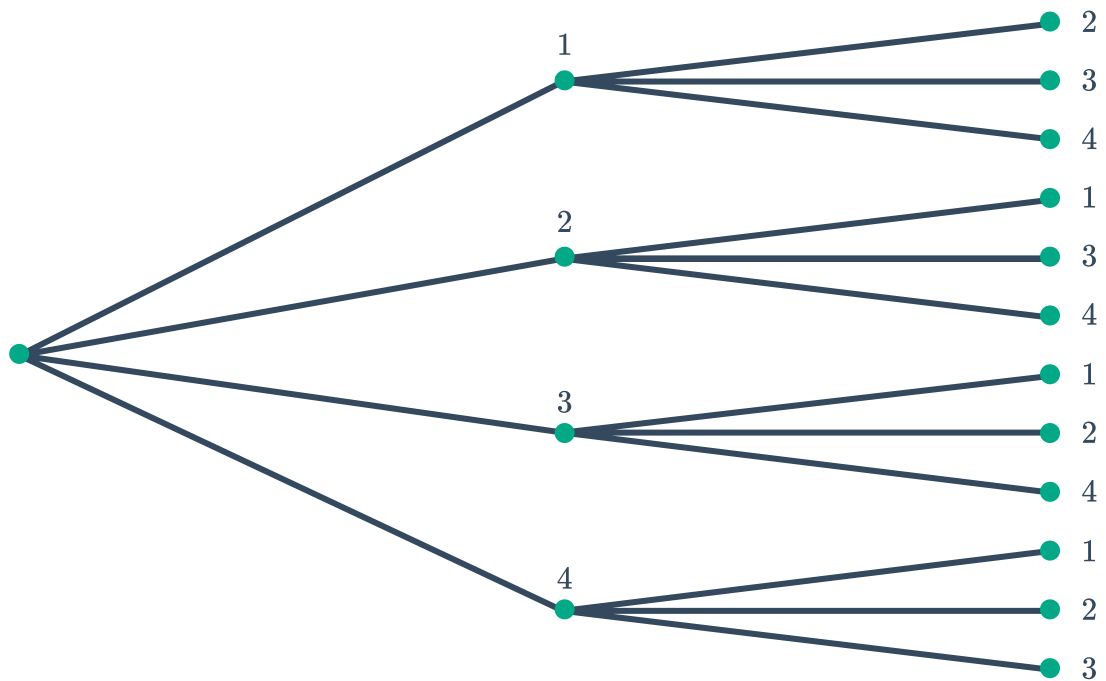


b 0.45

c 0.0855

28

a



b

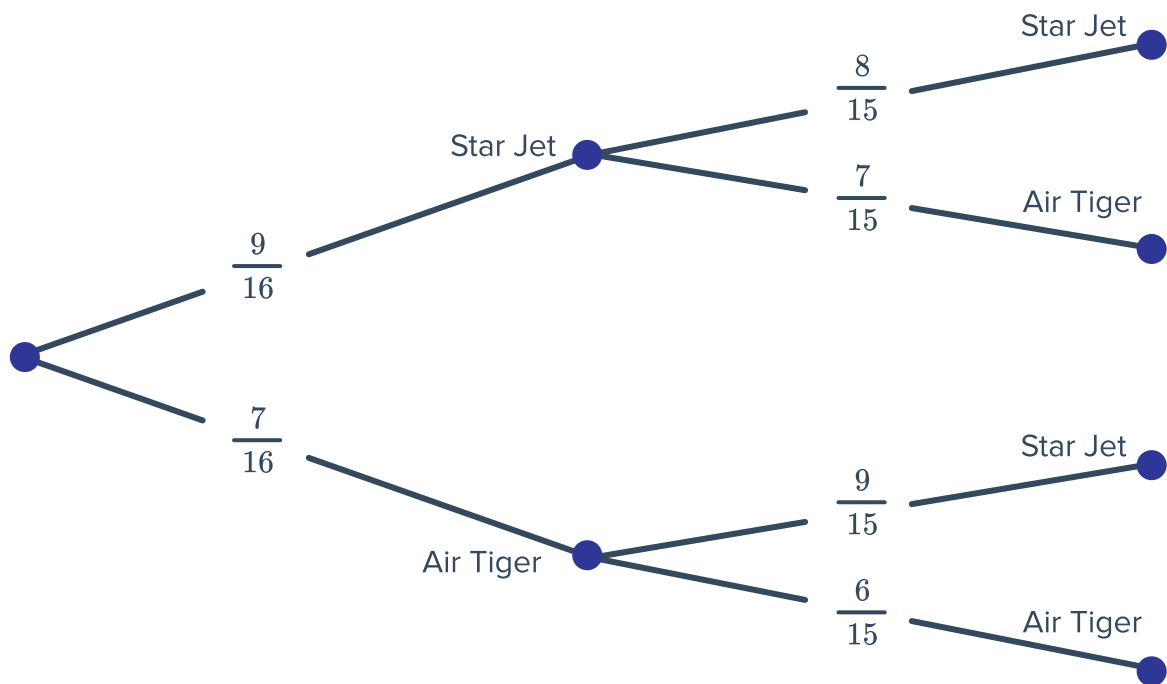
i $\frac{1}{2}$
v 0

ii 0
vi $\frac{1}{4}$

iii $\frac{1}{2}$
vii $\frac{1}{4}$

iv $\frac{1}{3}$

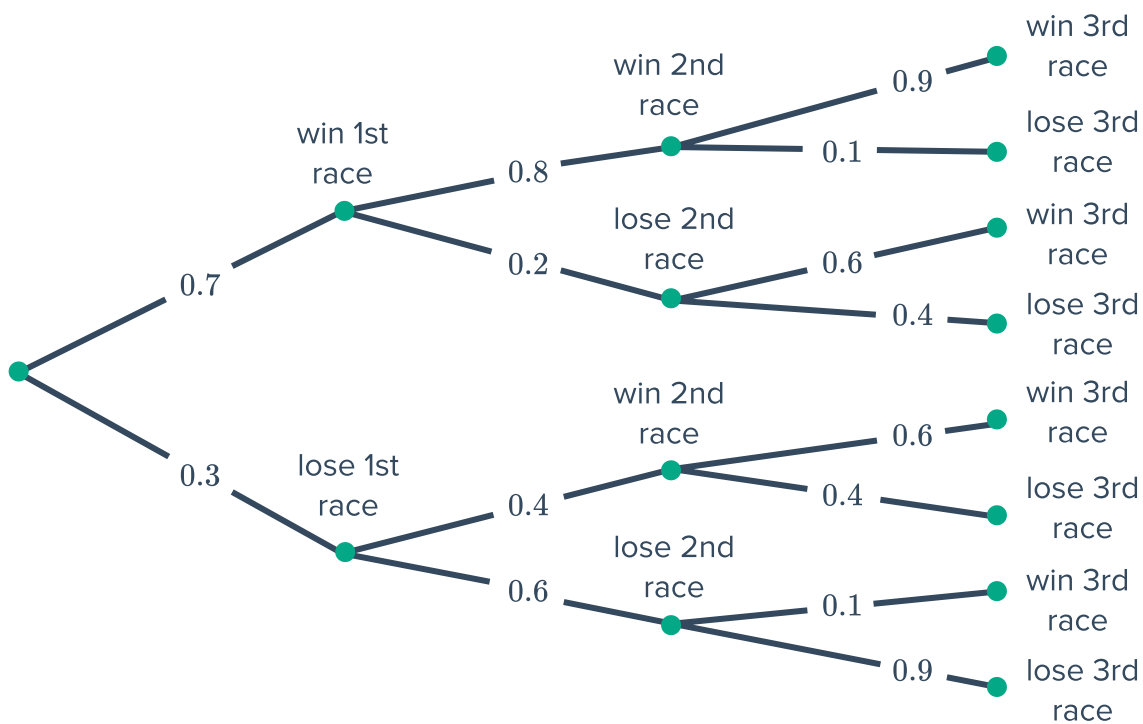
29
a



b
i $\frac{19}{40}$

ii $\frac{21}{40}$

30
a



b 0.504

c 0.678



Applications

31

a $\frac{1}{52}$

c $\frac{1}{52}$

e The probability is getting higher.

b $\frac{1}{52}$

d $\frac{1}{52}$

32

a $\frac{1}{55}$

b $\frac{1}{55}$

c Independent

33

a $\frac{1}{31}$

b Dependent

34

a $\frac{1}{3}$

b $\frac{1}{11}$

c $\frac{41}{55}$

35

a $\frac{1}{6}$

c $\frac{6325}{71\,253}$

e 25

b $\frac{25}{174}$

d The probability is getting higher.

36

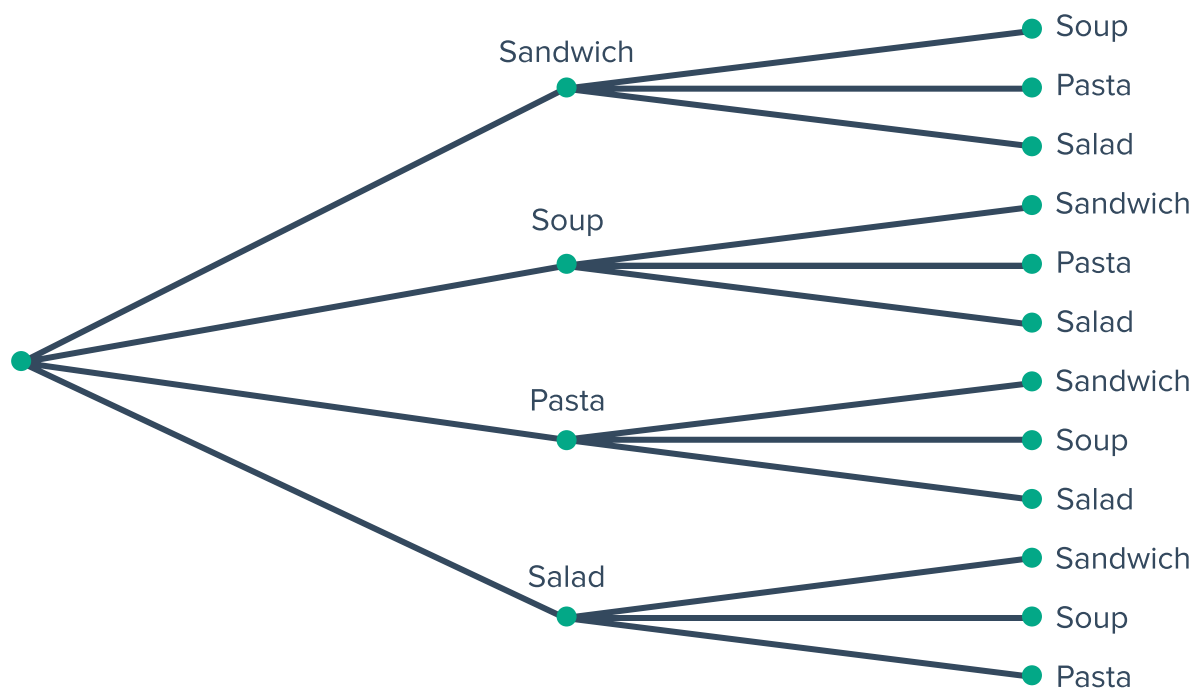
a 746, 764, 476, 467, 674, 647

c $\frac{1}{2}$

b $\frac{1}{2}$

37

a



b

i $\frac{1}{12}$

ii $\frac{1}{6}$

iii $\frac{1}{2}$

iv $\frac{1}{2}$

v $\frac{1}{12}$

38

a 2, 4, 6, 8, 10, 12, 14, 16, 18, 20

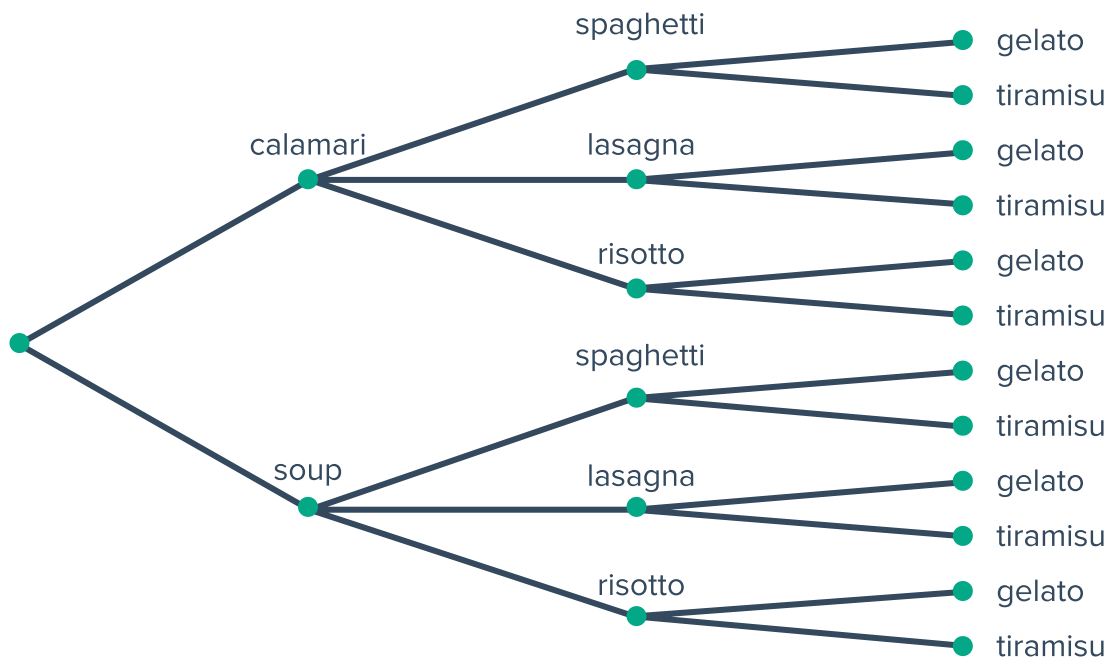
c 6

b 1, 2, 3, 4, 6, 8, 12, 16

d $\frac{3}{5}$

39

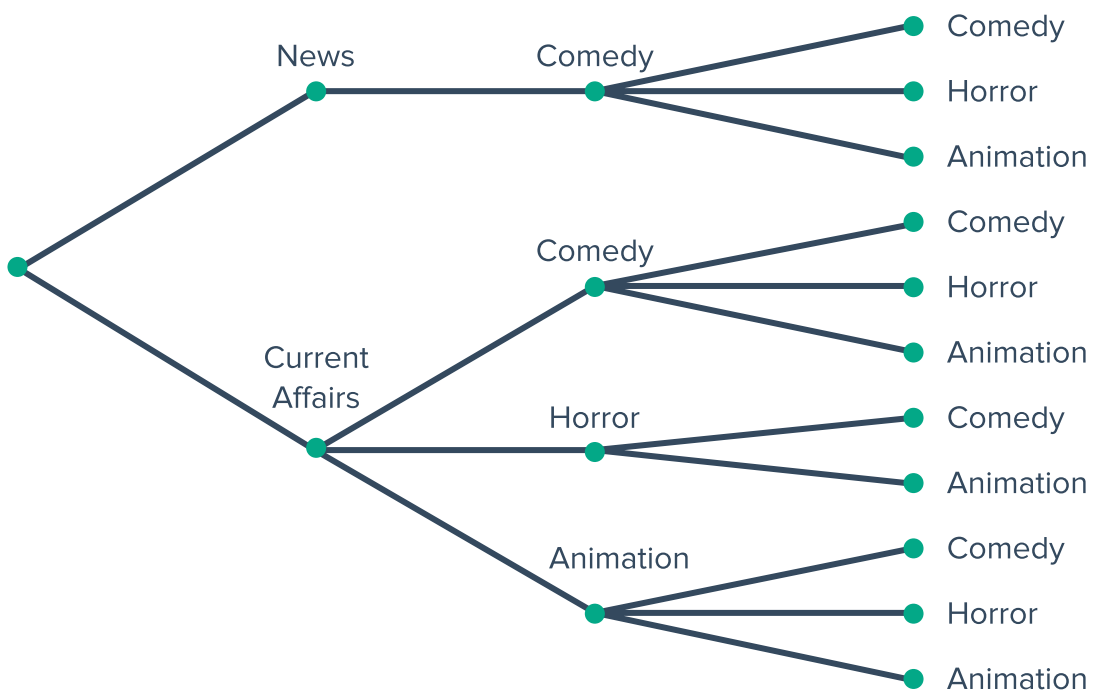
a



- b $\frac{1}{3}$
- c $\frac{1}{6}$
- d $\frac{3}{4}$

40

a



- b $\frac{1}{3}$

41

a



	Labor (next election)	Liberal (next election)	Greens (next election)	Total
Labor (last election)	26	41	3	70
Liberal (last election)	15	74	1	90
Greens (last election)	9	25	6	40
Total	50	140	10	200

b

$$\frac{1}{6}$$

c

$$\frac{2}{5}$$

d

$$\frac{23}{32}$$

42

a

$$\frac{1}{9}$$

b

$$\frac{5}{9}$$

c

$$\frac{1}{3}$$

43

a

i 260

ii

237

iii

111

iv

208

b

$$\frac{208}{319}$$

44

a

$$\frac{16}{69}$$

b

10

c

$$\frac{5}{13}$$

d

$$\frac{2}{3}$$

45

a

25

b

$$\frac{17}{87}$$

c

$$\frac{25}{42}$$

d

$$\frac{17}{27}$$

46

a

3

b

$$\frac{47}{50}$$

c

$$\frac{3}{10}$$

47



a

	Read the book	Didn't read the book	Total
Watched the movie	39	58	97
Didn't watch the movie	65	4	69
Total	104	62	166

b

i $\frac{65}{166}$

ii $\frac{39}{97}$

iii $\frac{29}{31}$

48

a

	Believe in Santa	Do not believe in Santa	Total
Believe in Tooth Fairy	31	10	41
Do not believe in Tooth Fairy	49	10	59
Total	80	20	100

b 38.75%

c $\frac{31}{90}$

d $\frac{10}{41}$

49

a

i $x = 426$

ii $y = 398$

iii $z = 154$

b $\frac{199}{489}$

c $\frac{77}{276}$

50

a

	Paperback	Not Paperback	Total
E-Reader	60	84	144
Not E-Reader	20	36	56
Total	80	120	200

b $\frac{21}{50}$

c 42%

51

a

i $x = 40$

ii $y = 400$

iii $z = 200$



b 360

c $\frac{10}{19}$